

ELEC 374 Digital Systems Engineering, W26

Lab CPU Phase 3 Demo, Report Submission, and Marking Guidelines

Some general guidelines for your Lab Demo, Lab Report submission, and marking rubrics are provided here:

1. Lab Demo:

- a) All members of a group must be present during their demo for each phase.
- b) Multiple (partial) lab demos for the same phase are NOT allowed, due to the large number of groups in the labs. When you are ready to demo your design, that would be your final demo for that phase. If you are behind, you have the opportunity to catch up during the later phases of the lab.
- c) The individual members of a group must demonstrate the understanding of their design. The TAs will check your Verilog/VHDL code for the Control Unit and its integration with the Datapath, and will ask the groups to explain how they have developed and integrated their Control Unit.
- d) The TAs will ask the groups to do a live simulation demo for a number of instructions in the test program. They will check the timing and correctness of the instructions according to their appropriate Control Sequence, as in previous phases. Make sure to include the required control signals and the register outputs in your waveforms, and also have a copy of the waveforms handy for your non-live demo to accelerate the demo process. Clearly mention what is working, what is not working, and discuss any outstanding issues or bonus designs.
- e) The TAs will check the opcode pattern, register numbers, and the contents of all registers and memory locations before and after simulation of the test program. This includes:
 - The contents of IR and the opcode pattern for all instructions, and if they comply with the CPU specification.
 - The contents of PC and if it is incremented by 1 for fetching the next instruction, or if the PC changes for the brmi and brpl branch instructions.
 - The contents of all registers including R0 to R15, HI, and LO registers.
 - The contents of memory locations 0x89 and 0xA3, and MDR and MAR registers.
- f) The TAs have been advised to spend around 25 minutes for each demo, given the number of groups we have in each lab section. So, please be patient with us until we get to your group.

2. Lab Report Submission

- a) For the contents of your lab report, please consult Section 4 of the Lab Phase 3 document.
- b) Lab Reports, in PDF format, must be submitted by 11:59pm to onQ on the same day of demo, and must comply with the lab demo. Otherwise, your lab report will not be accepted/marked. The marking TAs will check the date of the demo and the date the lab report was submitted. Submit only **one lab report** per group on onQ.
- c) Do not submit your lab report before the lab demo.
- d) Lab reports without a lab demo will not be accepted/marked.

- e) The lab marking will begin after the lab report submission deadline for each phase.
- f) Group members must abide by the academic integrity expectations, and include in their Lab Report a statement that says
 - *"We do hereby verify that this written lab report is our own work and contains our own original ideas, concepts, and designs. No portion of this report has been copied in whole or in part from another source, with the possible exception of properly referenced material".*

3. Lab Marking Rubric

The following marking scheme is used for the Lab Phase 3.

Verilog/VHDL code for the Control Unit and its integration with the Datpath in CPU (6), as well as correct simulation of instructions (6)	/12
Contents of IR for instructions, and if they comply with the CPU specification	/2.5
PC incrementing (1), or if PC changes for brmi and brpl branch instructions (2)	/3
Contents of all registers R0 to R15, HI, and LO (0.25 each) before and after simulation of the test program	/4.5
Contents of memory locations 0x89 and 0xA3 (2), as well as MDR and MAR (1) before and after simulation of the test program	/3
Phase 3 mark	/25
Note: bonus designs, outstanding issues, etc. Does the lab report comply with the lab demo? Bonus marks (5) will be determined after Lab Final Report is submitted at the end of the term.	